

Deep Learning: Systems and Responsibility

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ABSTRACT

Deep learning enables numerous applications across diverse areas. Data systems researchers are also increasingly experimenting with deep learning to enhance data systems performance. We present a tutorial on deep learning, highlighting the data systems nature of neural networks as well as research opportunities for advancements through data management techniques. We focus on three critical aspects: (1) classic design tradeoffs in neural networks which we can enrich through a systems and data management perspective, e.g., thinking critically about storage, data movement, and computation; (2) classic design problems in data systems which we can reconsider with neural networks as a viable design option, e.g., to replace or help system components that make complex decisions such as database optimizers; and (3) essential considerations for responsible application of neural networks in critical human-facing problems in society and how these also link to data management and performance considerations. While these are seemingly a diverse set of rich topics, they are strongly interconnected through data management, and their combination offers rich opportunities for future research.

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1 INTRODUCTION

Deep learning has seen an increasing amount of success. Deep neural networks are powerful computational models that can learn intricate and complex patterns directly from data [59]. By utilizing deep neural networks, researchers across several research communities can now solve problems that had evaded them for decades. These intricate computational models have come to power countless aspects of our society today. For instance, by applying deep learning models, researchers can localize and label objects within an image $3\times$ more accurately than the best classical methods [65]. In addition, deep neural networks can translate languages, help drive cars, and diagnose various diseases [12, 51, 71, 88]. Database systems researchers are also experimenting with neural networks and applying them to enhance database system performance [29, 54, 111].

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This widespread impact also raises critical concerns on designing and responsibly applying deep learning. These concerns arise because of three aspects of deep learning models: (a) their performance on any task is linked to the data we feed them with, (b) their training process is stochastic, and (c) using them consumes vast amounts of computing resources and energy.

This tutorial consists of three parts that blend deep learning with concepts and considerations that have to do with data management, data systems, and responsibility.

Part 1: A Systems Perspective on Deep Learning. Deep neural networks may consist of hundreds of compute-intensive layers of neurons trained through an iterative training procedure and deployed across heterogeneous devices. The primary challenge is to design methods that yield accurate deep neural networks while minimizing training time, inference time, and memory requirements.

In the first part of the tutorial, we provide the necessary background on neural networks from a systems perspective. We present various techniques to optimize for the core metrics looking holistically at both accuracy and system-level metrics. Since, in practice optimizing for one metric requires sacrificing another, we organize the various techniques in a framework that classifies them regarding how these techniques tradeoff between two or more metrics.

Part 2: Deep Learning for Data Systems. In the last five years, an increasing amount of research applies deep learning-based solutions to address a wide array of classical data systems problems – these range from efficient query processing to data exploration and visualization frameworks.

In the second part of the tutorial, we provide an overview of these techniques across various database system components. We explain how efficient data movement and computation techniques can make it viable to apply deep learning to even more components of a database system where we need both accuracy and efficient inference (e.g., the optimizer). We categorize approaches with respect to whether deep learning is used to enhance engineering/design decisions or replace an existing system component.

Part 3: Responsible Deep Learning. When deep learning is applied to user-facing applications, then there are new challenges to consider. For example, when we use deep learning to make decisions that can directly impact humans, it is important to understand how deep learning models made these decisions and why. In the third part of the tutorial, we explore responsibility across three dimensions: fairness, interpretability, and environmental impact. Again, we bring attention to challenges and opportunities for future research that are amenable to data management and systems solutions such as data provenance and efficient processing.

Audience. The tutorial is designed for an audience with a basic data management background (students, academics, researchers,